

Antibody-drug conjugates for advanced/recurrent high-grade endometrial cancer: potential targets, strategies and challenges

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Abstract

The incidence of endometrial cancer is on the rise, along with a concurrent increase in mortality. For advanced-stage endometrial cancer, standard treatment regimens include surgery, radiation, and cytotoxic chemotherapy; however, options are limited in the recurrent disease setting. Antibody-drug conjugates (ADCs) have emerged as a potent therapeutic option for patients who have experienced progression with chemotherapy, immunotherapy, or targeted therapy. Compared with conventional chemotherapy, ADCs deliver more potent cytotoxic payloads to the tumor by linking them to antibodies that target specific antigens preferentially expressed on malignant cells. Despite this advantage, resistance to ADCs has also emerged. Thus, it is important to understand the mechanisms of resistance and how they might be overcome to impact clinical outcomes. This review summarizes the current clinical application and ongoing development of ADCs in endometrial cancer, including evidence from preclinical studies and early-phase clinical trials, and discusses future challenges and directions.

Introduction

Endometrial cancer

Uterine cancer, which includes malignancies of the endometrium and myometrium, is the fourth most common cancer in women in the US (1). An estimated 68,270 new cases diagnosed and roughly 14,450 cancer deaths are projected to occur in 2026 (2). Unlike other gynecologic malignancies, the incidence of endometrial cancers (EC) is on the rise (3, 4). In 2024, EC surpassed ovarian cancer as the most lethal malignancy of the reproductive tract in the US. It is reported that the mortality difference may be due, in part, to a higher rate of aggressive histologic subtypes, especially among Black women compared with White women (5–8). Based on a state-transition microsimulation model of uterine cancer, it was predicted that the incidence and mortality of this disease will continue to increase over the next three decades (9). Furthermore, it predicted that Black women would continue to experience a disproportionate increase (9).

The most common histologic subtypes of EC include endometrioid (87.2%), serous (5.6%), carcinosarcoma (3.9%), clear cell (1.7%), and mixed type (1.7%). Low-grade endometrioid pathology makes up the better part of the new diagnoses and was historically referred to as Type I EC (10). High-grade EC, historically referred to as Type II EC, includes uterine serous carcinoma (USC), uterine carcinosarcoma (UCS), clear cell, grade 3 endometrioid, and undifferentiated (dedifferentiated) subtypes. While USC and UCS account for a small proportion of EC cases, they account for a significant share of EC mortality. Specifically, USC accounts for only about 10% of cases, yet is credited with 40% of deaths. The undifferentiated or dedifferentiated subtype is rare, with an estimated incidence of 1–9% (11).

In recent years, there has been some improvement in the prognosis for EC patients credited to advancements in developing treatment methods, including surgery, chemotherapy, radiotherapy, biologics, and immunotherapy (10, 12–15). The improvement in prognosis has been aided by the ever-mounting data accumulated from genomic and molecular profiling studies. Over time, we have gone from a more simplistic classification of EC (Type I and Type II) to an advanced molecular classification with four subtypes (DNA polymerase epsilon (POLE) ultramutated, microsatellite instability (MSI) high, microsatellite stable, copy number low (no specific molecular profile (NSMP)) and p53 mutated/DNA copy number high) (16, 17). These molecular subtypes are associated with distinct clinicopathologic features, prognostic outcomes, and potential therapeutic vulnerabilities. POLE-ultramutated tumors are characterized by pathogenic mutations in the exonuclease domain of POLE, an extremely high tumor mutational burden, and generally excellent clinical outcomes despite often high-grade histologic features. Microsatellite instability-high or mismatch repair-deficient tumors demonstrate defects in DNA mismatch repair, increased mutational burden, and prominent immune infiltration, making them particularly relevant to immune checkpoint inhibitor (ICI) -based treatment strategies. Copy-number low/microsatellite-stable tumors, also referred to as the NSMP, are a heterogeneous subset that often includes low-grade endometrioid carcinomas and typically have an intermediate prognosis; additional alterations, such as *CTNNB1* mutations, may further refine risk within this group (18). In contrast, copy-number high/p53-abnormal tumors are enriched for serous-like histology, frequent *TP53* alterations, extensive chromosomal instability, aggressive clinical behavior, and are generally associated with the poorest prognosis. Together, this molecular framework has improved risk stratification beyond histology alone and increasingly informs prognosis, adjuvant treatment decisions, and the development of subtype-specific therapeutic approaches. However, it is important to recognize that the bulk of the genomic and molecular profiling has been performed on treatment-naïve samples. It is also well accepted that treatment regimens can influence the genomic and molecular makeup of a tumor (19, 20). Regardless, the majority of recurrent EC will undergo a stepwise progression of treatment options, some of which are heavily influenced by upfront testing. This is due in part to limited access to post-treatment samples, unless sufficient biopsy material is recovered or tumor cells are isolated from fluid via paracentesis, thoracentesis, or pleural effusion.

Treatment options for recurrent EC include ICIs, such as pembrolizumab, dostarlimab, or durvalumab, in combination with carboplatin and paclitaxel (13–15). For patients with mismatch repair proficient (pMMR) tumors, lenvatinib plus pembrolizumab remains an established later-line option, particularly after platinum-based chemotherapy. Notably, an exploratory subsequent-therapy analysis of NRG-GY018 suggested that this combination may retain clinical activity even after prior exposure to pembrolizumab with carboplatin and paclitaxel, although prospective data defining its efficacy in this setting remain limited (12, 21, 22). Treatment selection for recurrent EC is individualized according

to prior treatment exposure, mismatch repair/microsatellite instability (MSI) status, histology, hormone receptor expression, Human epidermal growth factor receptor 2 (HER2) status, and the distribution of recurrent disease. In patients with deficient mismatch repair (dMMR)/MSI-High (MSI-H) recurrent disease who have not received prior immunotherapy, single-agent pembrolizumab or dostarlimab may also be considered after platinum-based therapy (23, 24). HER2-directed therapy is another treatment strategy for patients with HER2-positive serous or p53-abnormal tumors; trastuzumab may be added to carboplatin and paclitaxel, and trastuzumab deruxtecan (T-DXd) may be considered for previously treated, unresectable, or metastatic HER2-positive tumors (25). In addition, exploratory biomarker analyses of the GOG-86P trial suggested that patients with TP53-mutated or p53-overexpressing tumors may derive particular benefit from the addition of bevacizumab to carboplatin and paclitaxel. However, these findings remain hypothesis-generating and require prospective validation before this regimen can be regarded as a molecularly selected standard of care (26, 27). Hormonal therapy remains an option for patients with low-grade, hormone receptor-positive endometrioid carcinoma, particularly in indolent or low-volume disease (28, 29). In addition, local therapies, including surgery or radiation therapy, may be appropriate for isolated or locoregional recurrence, especially in patients who have not previously received radiation (30, 31).

Despite these advancements in matching therapeutic options with patients with different hormone receptor status, MMR status, HER2 expression, and p53 mutations, subsequent treatment options for patients who experience disease progression are limited by low efficacy and high toxicity. Continued development of novel treatment strategies is critical to enhance efficacy and ultimately patient prognosis. In response, the field is focusing heavily on antibody-drug conjugates (ADCs). To date, no disease-specific approvals of ADCs have been granted for EC.

Antibody drug conjugates

ADCs are a rapidly evolving therapeutic modality designed to selectively deliver potent cytotoxic payloads via tumor-specific target antigens, thereby minimizing off-target toxicities that would otherwise be associated with systemic delivery of more potent payloads. Fundamentally, an ADC consists of a target-specific monoclonal antibody conjugated to a cytotoxic payload via a linker that remains stable in circulation, only releasing its payload once it reaches its target and is taken up by the cell (32) (Fig. 1.). Another important design characteristic is the drug-to-antibody ratio (DAR), which represents the average number of payload molecules conjugated to each antibody. DAR can influence ADC potency, pharmacokinetics, stability, clearance, efficacy, and toxicity; although increasing the DAR can increase payload delivery, excessively high drug loading may increase hydrophobicity, accelerate systemic clearance, and narrow the therapeutic window (33, 34).

After binding to its target antigen on the tumor cell surface, the ADC-antigen complex is internalized, typically through receptor-mediated endocytosis, and trafficked through endosomal compartments to the lysosome. Within the lysosome, the payload is released through proteolytic degradation of the antibody and/or cleavage of the linker, depending on the linker design. The released cytotoxic agent then reaches its intracellular target and induces tumor cell death, most commonly by disrupting microtubule dynamics or inhibiting DNA replication and repair (32, 35). The efficacy of an ADC is therefore influenced not only by target antigen expression, but also by the efficiency of internalization, intracellular trafficking, linker cleavage, and payload release (Fig. 2.).

Some ADCs can additionally exert a bystander effect. In this process, membrane-permeable payloads released from antigen-positive tumor cells diffuse into neighboring cells and induce cytotoxicity, including in cells with low or absent target-antigen expression. This effect may be particularly important in tumors with heterogeneous antigen expression. The magnitude of the bystander effect depends largely on the physicochemical properties of the payload, such as its hydrophobicity and the use of a cleavable linker (36). Although the bystander effect may broaden antitumor activity, it may also contribute to toxicity in adjacent normal tissues.

Antibody drug targets in EC

To provide a consistent and clinically relevant overview, the main text focuses on ADCs with either reported EC-specific clinical efficacy data or progression to an EC-focused phase III trial. Agents evaluated only in early-phase basket studies, those without reported EC-specific outcomes, and preclinical programs are summarized in the corresponding tables.

HER2

HER2 function

HER2 is a transmembrane protein that is a member of the epidermal growth factor (erythroblastic oncogene B (ERBB)) family (37). Once activated, it signals cell proliferation, survival, and differentiation (38). Unlike other ERBB family members, HER2 does not directly bind ligand. Its activation is stimulated by homodimerization with itself or heterodimerization with other epidermal growth factor receptors (HER1, 3, or 4) (37). When levels of HER2 are aberrantly elevated, whether by increased copy number or other alterations, they can lead to the development or progression of cancer (39).

HER2 expression and role in EC

Overexpression of HER2 is associated with high-grade, aggressive EC, which carry a higher risk of recurrence and poor overall survival (OS) rates (17, 40). Numerous studies show that over 90% of HER2-amplified endometrial tumors harbor concurrent *TP53* mutations (17, 41, 42). Despite HER2 overexpression, patients treated with trastuzumab (Herceptin), a monoclonal antibody against HER2, as a single agent had limited success. Studies using preclinical models of EC suggested that trastuzumab combined with lapatinib was more effective than trastuzumab monotherapy (43). Subsequent clinical studies showed that combining trastuzumab with chemotherapy extended the progression-free survival (PFS) for patients with advanced or recurrent HER2-positive EC (44). As such, HER2 is being actively targeted for ADCs in EC.

HER2 ADCs in clinical trials

T-DXd is an ADC comprising an immunoglobulin G1 monoclonal antibody targeting HER2, conjugated through a cleavable tetrapeptide-based linker to a membrane-permeable topoisomerase I inhibitor (TOP01i) payload, with a DAR of approximately 8. In the initial DESTINY-PanTumor02 study (NCT04482309), T-DXd demonstrated clinically meaningful response rates in pretreated patients with HER2-expressing solid tumors, with benefit observed in endometrial, cervical, and ovarian cancers (45). This study demonstrated that T-DXd was active in HER2 immunohistochemistry (IHC) 2+/3 + EC, with a response rate of 57.5% (95% CI, 40.9–73.0), a median PFS of 11.1 months (95% CI, 7.1–not reached), and an OS of 26 months (95% CI, 12.8–not reached) (45). Two phase III trials have since been initiated to validate these findings. DESTINY-Endometrial01 (NCT06989112) is evaluating T-DXd in combination with either pembrolizumab or rilvegostomig as first-line therapy for HER2-expressing (IHC 2 + or 3+), pMMR primary advanced or recurrent EC. Rilvegostomig is a bispecific immune checkpoint antibody that simultaneously binds and inhibits programmed cell death protein 1 (PD-1) and T-cell immunoreceptor with immunoglobulin and ITIM domains (TIGIT), thereby restoring antitumor T-cell activity through dual checkpoint blockade. DESTINY-Endometrial02 (NCT07022483) is evaluating T-DXd, with or without radiotherapy, versus standard-of-care chemotherapy with or without radiotherapy, as adjuvant treatment after surgery for HER2-expressing (IHC 2 + or 3+) EC.

Trastuzumab Pamirtecán (BNT323 or DB-1303) is another ADC in clinical trials that targets HER2 and uses a TOP01i payload distinct from T-DXd. In the phase 1/2a study NCT05150691, BNT323 demonstrated antitumor activity in previously treated patients with advanced or metastatic HER2-expressing EC. Among patients with centrally confirmed HER2-expressing tumors, the confirmed objective response rate (ORR) was 47.9% (95% CI, 37.6–58.4), the median duration of response was 11.1 months (95% CI, 9.0–18.3), and the median PFS was 8.1 months (95% CI, 5.5–11.8). In the prespecified subgroup previously treated with an ICI, the confirmed ORR was 49.3% (95% CI, 37.4–61.3), with a median duration of response of 9.9 months (95% CI, 7.0–not evaluable) and a median PFS of 6.8 months (95% CI, 5.4–11.0). Activity was observed across HER2 expression levels, with centrally assessed ORRs of 34.5%, 44.2%, and 70.8% in tumors with HER2 IHC 1+, 2+, and 3 + expression, respectively. Grade 3 or higher treatment-related adverse events occurred in 46.9% of patients, and interstitial lung disease or pneumonitis remained an important toxicity requiring proactive monitoring (46). BNT323 is also being evaluated in the Fern-EC-01 (NCT06340568), an open-label, randomized, multi-site, Phase III study. It is designed to determine the efficacy and safety of BNT323 compared with the investigator's choice of single-agent chemotherapy in previously treated participants with recurrent EC who have progressed on at least one line of platinum-based therapy and ICI (Cohort 1). For this study, HER2 is assessed centrally using a protocol-defined IHC scoring algorithm; however, publicly available information does not specify whether breast or gastric cancer HER2 scoring criteria are applied. HER2 expression is defined as a 1+, 2+, or 3 + score based on IHC. Participants with recurrent EC determined to have a HER2 IHC 3 + score will be enrolled in the BNT323 monotherapy arm (Cohort 2) to further evaluate the efficacy and safety of BNT323.

Additional HER2-targeted ADCs and earlier-stage studies are summarized in Table 1.

Table 1
HER2-directed antibody-drug conjugates evaluated in endometrial cancer

ADC	Payload	Linker	DAR	Trial (NCT#)	Study population	Phase
Trastuzumab deruxtecan (T-DXd; DS-8201a)	DXd an exatecan-derived topoisomerase I inhibitor	Enzyme-cleavable tetrapeptide	~ 8	DESTINY-PanTumor02 (NCT04482309)	Basket (solid tumors)	II
				DESTINY-Endometrial01 (NCT06989112)	EC-focused	III
				DESTINY-Endometrial02 (NCT07022483)	EC-focused	III
				DESTINY-PANTUMOUR04 (DP-04) (NCT07124000)	Basket (solid tumors)	Observational
				NCT04639219	Basket (solid tumors)	II
				ESPERANZA (NCT06973161)	EC-focused	Observational
				DASH (NCT04704661)	Basket (solid tumors)	I
				PETRA (NCT04644068)	Basket (solid tumors)	I/II
Trastuzumab pamirtecán (BNT323/DB-1303)	P1003 a proprietary topoisomerase I inhibitor	Enzymatically cleavable tetrapeptide linker	~ 8	NCT05150691	Basket (solid tumors)	I/IIa
				BNT323-01/Fern-EC-01 (NCT06340568)	EC-focused	III
vic-trastuzumab duocarmazine (SYD985)	Seco-DUBA a duocarmycin-derived DNA-alkylating agent	Protease-cleavable valine-citrulline linker	~ 2.8	NCT04205630	EC-focused	II
				NCT04602117	EC-focused	Withdrawn
Trastuzumab rezetecan (SHR-A1811)	Rezetecan/SHR169265 a topoisomerase I inhibitor	Enzyme-cleavable linker	~ 6	NCT05896020	EC-focused	II
Ado-trastuzumab emtansine (T-DM1)	DM1/emtansine a maytansinoid microtubule inhibitor	Non-cleavable MCC thioether linker	~ 3.5	NCT04486352	EC-focused	Ib/II
				NCI-MATCH, Subprotocol Q (NCT02465060)	Basket (molecularly selected tumors)	II
Disitamab vedotin (RC48)	Monomethyl auristatin E (MMAE) a microtubule-disrupting agent	Protease-cleavable valine-citrulline linker	~ 4	NCT06003231	EC-focused	II

Folate Receptor Alpha (FR α)

FR α function

Folate receptor α (FR α) is a cell membrane protein that binds and transports folate (Vitamin B9) via endocytosis into the cell (47). FR α can mediate DNA synthesis, regulate gene expression, induce cell proliferation, and support brain function and embryonic development (48–50). Its presence and activity are limited in normal healthy cells, but FR α levels are elevated in multiple epithelial malignancies, making it a candidate for targeted therapy (47, 51).

FR α expression and role in EC

FR α expression has been reported in several histologic subtypes of EC, particularly serous carcinoma, high-grade endometrioid carcinoma, and carcinosarcoma (52). Earlier immunohistochemical studies identified FR α expression in a substantial subset of high-risk EC, while more recent analyses using the clinically validated VENTANA FOLR1 RxDx assay have reported a lower prevalence, with positivity concentrated predominantly in serous tumors (53). However, the reported prevalence varies substantially among studies, ranging from approximately 11%, in endometrial serous carcinomas evaluated using the stringent VENTANA FOLR1 RxDx threshold, to 87%–100%, in UCS assessed using less restrictive immunohistochemical criteria. This reflects differences in patient populations, antibodies, scoring systems, and thresholds used to define FR α positivity (53–55). Although FR α expression is enriched in aggressive histologic subtypes, its independent prognostic significance in EC remains uncertain. Available analyses have not consistently demonstrated an association between FR α expression and PFS or OS, and studies in UCS have similarly found no significant relationship between FR α expression and clinicopathologic features or prognosis (55). In UCS, FR α expression appears to be more frequent and intense in the carcinomatous component than in the sarcomatous component, supporting the potential use of FR α -directed therapies in selected patients (54).

FR α expression has also been reported to increase from normal endometrium and endometrial hyperplasia to endometrioid carcinoma, suggesting a possible role in endometrial carcinogenesis and tumor progression (56). Functionally, its cell-surface localization and reported internalization. Ligand or antibody binding supports the development of FR α -directed drug-delivery strategies. Nevertheless, FR α expression is variable within and between tumors, and antigen expression alone may not fully predict clinical response (57, 58). Standardized testing and assessment of recurrent or metastatic tissue will therefore be important for identifying patients most likely to benefit from FR α -directed ADCs.

FR α ADCs in clinical trials

Mirvetuximab soravtansine (MIRV) is an FR α targeting ADC (59). It is conjugated via a cleavable linker to a DM4 (ravtansine) payload, a maytansinoid, with an average DAR of approximately 3.5 (60). MIRV binds to FR α with high affinity and specificity and is internalized via endocytosis. The vesicle travels to the lysosome, where the linker is cleaved, releasing the intracellular DM4 payload (59). DM4 suppresses microtubule dynamic instability, which triggers cell cycle arrest and cell death. Clinical activity has also been reported in a single-arm phase II study of MIRV combined with pembrolizumab in patients with recurrent or persistent FR α -expressing serous EC. Eighteen patients received treatment, and the confirmed ORR was 28% (95% CI, 10–53%), including one complete response and four partial responses. The estimated 6-month PFS rate was 24.4% (95% CI, 7.7–46.1%). Grade 3 treatment-related adverse events were uncommon, and no grade 4 or higher treatment-related toxicities were reported. However, the study closed before reaching its planned sample size due to recruitment constraints, and these findings should therefore be considered preliminary (58, 61).

Rinatabart sesutecan (Rina-S; PRO1184; GEN1184) is another FR α -targeting ADC that incorporates an exatecan-derived TOP01i payload through a protease-cleavable, highly hydrophilic linker, with a DAR of approximately 8. Rina-S is being evaluated in the phase I/II RAINFOL-01 basket study (NCT05579366), which includes patients with endometrial cancer. Its development has advanced to the randomized phase III RAINFOL-03 trial (NCT07166094), which compares Rina-S with the investigator's choice of treatment in patients with endometrial cancer previously treated with platinum-based chemotherapy and PD-(L)1-directed therapy. Clinical efficacy results specifically in endometrial cancer have not yet been reported.

Additional FR α -targeting ADCs and earlier-stage clinical studies are summarized in Table 2.

Table 2
FRα-directed antibody-drug conjugates evaluated in endometrial cancer

ADC	Payload	Linker	DAR	Trial (NCT#)	Study population	Phase
Mirvetuximab soravtansine (MIRV; IMGN853)	DM4 a maytansinoid microtubule inhibitor	Cleavable sulfo-SPDB disulfide linker	~ 3.4–3.5	NCT03832361	EC-focused	II
				NCT03835819	EC-focused	II
				NCT03836157	EC-focused	II
Luveltamab tazevibulin (STRO-002; Luvelta)	SC209 a 3-aminophenyl hemiasterlin microtubule inhibitor	Protease/cathepsin-B-cleavable valine-citrulline linker (SC239 linker-payload)	4	NCT03748186	Basket (gynecologic cancers)	I
Rinatabart sesutecan (Rina-S; PRO1184; GEN1184)	Exatecan a topoisomerase I inhibitor	Novel protease-cleavable, highly hydrophilic linker	8	RAINFOL-01 (NCT05579366)	Basket (solid tumors)	I/II
				RAINFOL-03 (NCT07166094)	EC-focused	III
Opugotamig olatansine (IMGN151)	DM21 a maytansinoid microtubule inhibitor	Stable cleavable peptide linker	~ 3.5	NCT05527184	Basket (solid tumors)	I
				NCT07024784	EC-focused	Ib
Farletuzumab ecteribulin (MORAb-202)	Eribulina a microtubule-dynamics inhibitor	Cathepsin-B-cleavable Mal-PEG ₂ -valine-citrulline-PABC linker	4	NCT04300556	Basket (solid tumors)	I/II
AMT-151	Duostatin-5 (DUO-5) a microtubule-disrupting payload	Cleavable linker	ND	NCT05498597	Basket (solid tumors)	I

TROP2

TROP2 function

Trophoblast Cell Surface Antigen 2 (TROP2) is a transmembrane glycoprotein. Its cellular functions involve regulating intracellular calcium signaling, cell cycle progression, cell-to-cell adhesion, and motility (62–64). Because its expression is overexpressed in many solid tumors compared with normal healthy tissues, TROP2 is currently being targeted by ADCs for the treatment of breast, lung, gastrointestinal, and ovarian cancers, among others (65, 66).

TROP2 expression and role in EC

Investigations into TROP2 as a potential driver of aggressive features of EC are not new. An early report by Bignotti et al. showed that TROP2 expression was detectable by IHC in 126 out of 131 (96.2%) endometrial endometrioid tumor samples on a tissue microarray (67). These tumor tissue samples showed a higher level of TROP2 positivity than normal endometrial tissue samples ($p = 0.001$). Furthermore, grade 3 endometrioid EC showed stronger TROP2 immunostaining than grade 1 tumor samples ($p = 0.01$). In a subsequent analysis with clinical outcome data available for 103 patients, strong TROP2 expression was associated with higher tumor grade and cervical involvement. TROP2 overexpression was significantly associated with shorter disease-free survival in univariate analysis (HR, 3.36; 95% CI, 1.26–8.92; $P = 0.01$) and remained an independent predictor of disease recurrence in multivariable analysis (HR, 2.82; 95% CI, 1.05–7.58; $P = 0.04$), together with advanced FIGO stage (HR, 5.36; 95% CI, 2.04–14.08; $P < 0.001$). Associations with PFS and OS showed only borderline significance in univariate analysis (PFS: HR, 2.80; 95% CI, 0.99–7.87; $P = 0.05$; OS: HR, 2.60; 95% CI, 0.94–7.19; $P = 0.06$) and were not significant after multivariable adjustment, indicating that the prognostic evidence is strongest for disease-free survival in endometrioid EC (68).

TROP2 ADCs in clinical trials

Several TROP2-targeted ADCs are currently being evaluated in clinical trials for EC (Table 3).

Sacituzumab govitecan (SG) consists of an anti-TROP2 monoclonal antibody conjugated to SN-38, the active metabolite of irinotecan, through a hydrolyzable linker, with a DAR of approximately 7.6. In the EC cohort of the phase II TROPiCS-03 basket trial (NCT03964727), SG demonstrated an ORR of 22%, with a median duration of response of 8.8 months and a median PFS of 4.8 months in patients with heavily pretreated EC (69). SG is also currently being compared with the physician's choice of chemotherapy in the randomized phase III ASCENT-GYN-01 trial (NCT06486441) in patients who have previously received platinum-based chemotherapy and a PD-(L)1 inhibitor.

Sacituzumab tirumotecan (sac-TMT; MK-2870) is another TROP2-targeted ADC that incorporates a belotecan-derived TOPO1i payload, with a DAR of approximately 7.4. The randomized phase III TroFuse-005 trial (NCT06132958) is comparing sac-TMT with the physician's choice of chemotherapy in patients with advanced or recurrent EC who had previously received platinum-based chemotherapy and immunotherapy (70). Sac-TMT is also being evaluated in the phase III TroFuse-033 trial (NCT06952504) as part of first-line treatment for patients with advanced or recurrent EC.

Additional TROP2-targeted ADCs and earlier-stage clinical studies are summarized in Table 3.

Table 3
TROP2-directed antibody-drug conjugates evaluated in endometrial cancer

ADC	Payload	Linker	DAR	Trial (NCT#)	Study population	Phase
Sacituzumab govitecan (SG; IMMU-132)	SN-38 the active metabolite of irinotecan and a topoisomerase I inhibitor	Hydrolysable CL2A carbonate linker	~ 7.6	IMMU-132-01 (NCT01631552)	Basket (solid tumors)	I/II
				TROPiCS-03 (NCT03964727)	Basket (solid tumors)	II
				GOG-3023 (NCT04251416)	EC-focused	II
				NCT06040970	EC-focused	I
				ASCENT-GYN-01/GOG-3104/ENGOT-en26 (NCT06486441)	EC-focused	III
				NCT06865677	Basket (gynecologic cancers)	II
Sacituzumab tirumotecan (Sac-TMT; MK-2870; SKB264)	KL610023 a belotecan-derived topoisomerase I inhibitor	2-Methylsulfonyl pyrimidine–CL2A–carbonate linker	~ 7.4	NCT05867251	Basket (solid tumors)	I/II
				KL264-01 (NCT04152499)	Basket (solid tumors)	I/II
				NCT05642780	EC-focused	II
				TroFuse-005/MK-2870-005 (NCT06132958)	EC-focused	III
				TroFuse-033/MK-2870-033 (NCT06952504)	EC-focused	III
Datopotamab deruxtecan (Dato-DXd; DS-1062a)	DXd an exatecan-derived topoisomerase I inhibitor	Enzyme-cleavable tetrapeptide-based linker	4	TROPION-PanTumor03 (NCT05489211)	Basket (solid tumors)	II
				PETRA: Dato-DXd plus AZD5305 (NCT04644068)	Basket (solid tumors)	I/IIa

B7-H4

B7-H4 function

B7 homolog 4 (B7-H4) is a member of the B7 family of immune checkpoint proteins, which includes PD-1 and programmed death-ligand 1 (PD-L1). Its role is to act as a secondary signal to stimulate or suppress T cell proliferation, cytokine production, and overall effector function. B7-H4, therefore, mediates tumor immune evasion and is inversely correlated with T-cell tumor infiltration (71–73). Overexpression of B7-H4 has been reported in many tumor types (74), including ovarian, lung, breast, pancreatic, and renal cell cancer (75–79), whereas it is not expressed in normal human tissues (80), making it a candidate for ADC-targeted therapies.

B7-H4 expression and role in EC

We and others have demonstrated that B7-H4 is highly expressed in EC. Specifically, our study compared B7-H4 IHC staining in 40 low-grade and 30 high-grade endometrioid tumors. We found B7-H4 expression was ubiquitous across tumor grades, though we observed high-grade EC tumors were more likely to have elevated PD-L1 and B7-H4 simultaneously (72). Another study found that B7-H7 staining increased from normal endometrial tissue to atypical endometrial hyperplasia to invasive endometrioid adenocarcinoma (81). Additionally, the study showed high risk tumors (age > 50, all grades, all stages) had a higher proportion of B7-H4 positive tumor cells compared to low risk tumors (age < 50, grade 1, stage 1) (81). Overexpression of B7-H4 in EC tumors compared to normal cells has been consistently demonstrated in additional studies (73, 80, 82).

B7-H4 ADCs in clinical trials

Several B7-H4-targeted ADCs are currently being evaluated in clinical trials for EC (Table 4).

Mocertatug rezetecan (Mo-Rez / GSK5733584) is a B7-H4 ADC that utilizes an exatecan-derived TOP01i with a DAR of 6. A Study to Evaluate the Safety, Tolerability, Pharmacokinetics and Clinical Activity of GSK5733584 for Injection in Participants with Advanced Solid Tumors (BEHOLD-1), an early-phase clinical trial (NCT06431594), enrolled adult patients with histologically confirmed advanced solid tumors (83, 84). Among evaluable patients with EC, Mo-Rez demonstrated an ORR of 67% (95% CI, 35%–90%) (85). Mo-Rez has subsequently advanced to the EC-focused phase III BEHOLD-Endometrial01 trial (NCT07286331). Additional B7-H4-targeted ADCs are summarized in Table 4.

Table 4
B7-H4-directed antibody-drug conjugates evaluated in endometrial cancer

ADC	Payload	Linker	DAR	Trial (NCT#)	Study population	Phase
Mocertatug rezetecan	SHR-9265/GSK5757810A	Protease-cleavable linker	6	BEHOLD-1 (NCT06431594)	Basket (solid tumors; EC/PROC expansion)	I II III
(Mo-Rez; GSK5733584; HS-20089)	an exatecan-derived topoisomerase I inhibitor			HS-20089 (NCT06014190)	Basket (solid tumors)	II
				BEHOLD-Endometrial01 (NCT07286331)	EC-focused	III
Emiltatug ledadotin	Auristatin F-hydroxypropylamide (AF-HPA)	Cleavable ester-containing Dolasynthen linker platform	6	NCT05377996	Basket (solid tumors)	I
(Emi-Le; XMT-1660)	a microtubule-disrupting auristatin payload					
Puxitatug samrotecan	AZ14170133	Protease-cleavable Val-Ala-PEG8 linker	8	BLUESTAR (NCT05123482)	Basket (solid tumors)	I/IIa
(Puxi-Sam; AZD8205)	a membrane-permeable topoisomerase I inhibitor			BLUESTAR-Endometrial01/GOG-3110/ENGOT-en28 (NCT07044336)	EC-focused	III
Felmetatug vedotin	Monomethyl auristatin E (MMAE)	Protease-cleavable maleimidocaproyl-valine-citrulline linker	~ 4	NCT05194072	Basket (solid tumors)	I
(SGN-B7H4V; PF-08046048)	a microtubule-disrupting agent					

ADC's in early clinical development

The following section describes antigens in early clinical development for EC.

MUC16 (CA-125)

MUC16 function

Mucin 16 (MUC16) is a large, heavily glycosylated transmembrane mucin expressed on the apical surface of several epithelial tissues (86). Its large extracellular domain contributes to epithelial protection and lubrication under physiological conditions and can be proteolytically shed, releasing the CA-125 antigen into the circulation. In cancer, MUC16 has been implicated in cell adhesion, metastatic dissemination, immune evasion, and activation of oncogenic signaling pathways. Interactions between MUC16 and mesothelin may facilitate tumor cell attachment to mesothelial surfaces, whereas its extensive, glycosylated extracellular domain may provide a physical barrier to immune cell recognition (86). However, many of these functions have been characterized primarily in ovarian and other epithelial cancers, and their relevance to EC remains incompletely defined.

MUC16 expression and role in EC

MUC16 expression in EC is heterogeneous and varies according to histologic subtype and tumor differentiation (87). In an immunohistochemical analysis of 485 EC patient tumors, CA-125/MUC16 expression was detected in 368 tumors (75.9%), although the extent and intensity of expression varied among cases (88). Accordingly, MUC16 is not uniformly expressed across all EC subtypes and is not routinely used as a diagnostic tissue marker for EC. Serum CA-125 is likewise not recommended as a general diagnostic or screening biomarker for EC; however, elevated serum CA-125 levels in patients with EC have also been associated with adverse clinicopathologic features, including advanced-stage disease, deep myometrial invasion, lymphovascular or lymph-node involvement, and extrauterine spread (89–91). Thus, serum CA-125 may provide supplementary information regarding disease burden and prognosis in selected patients, particularly those with advanced or high-risk disease (89–91). However, MUC16 is currently being developed as a potential target for ADCs in EC.

MUC16 ADCs in clinical trials (Table 5)

MUC16-targeted ADCs remain in early clinical development for EC. HWK-016 is an ADC that specifically targets the non-shed, membrane-bound region of MUC16. The phase I/Ib trial (HWK-016-101; NCT07470853) is a multicenter, open-label, first-in-human study evaluating HWK-016 in patients with advanced or metastatic solid tumors, including EC.

NaPi2b

NaPi2b function

Sodium-dependent phosphate transport protein 2b (NaPi2b) is a transmembrane sodium-dependent phosphate transporter encoded by the gene *SLC34A2* and plays a role in the maintenance of phosphate metabolism (92). Under physiological conditions, NaPi2b mediates sodium-dependent phosphate uptake across epithelial cell membranes and contributes to epithelial transport and cellular phosphate availability(93). Its cell-surface localization and differential expression between selected tumors and normal tissues make NaPi2b an attractive target for antibody-based drug delivery (94). In malignant cells, aberrant NaPi2b expression may reflect epithelial lineage and altered phosphate homeostasis; however, its specific functional contribution to EC development and progression remains incompletely understood(95).

NaPi2b expression and role in EC

By utilizing publicly available tumor gene expression data from cBioPortal TCGA and Genie datasets, Vlasenkova et. al. identified uterine tumor tissues had higher expression of *SLC34A2* compared to healthy control tissues (95). At the protein level, the Human Protein Atlas shows high expression of NaPi2b in endometrial tumors using two independent antibodies, HPA037989 and HPA066474 (96). Internal IHC data was presented that identified approximately 80% of EC tumors were positive for NaPi2b (97).

NaPi2b ADCs in clinical trials (Table 5)

As such, there are preliminary studies working to develop NaPi2b targeted ADCs for EC. YL205 is a NaPi2b targeting ADC. It comprises a humanized monoclonal antibody, conjugated to the linker and payload via the proprietary TMALIN (Tumor Microenvironment Activable LINKer-payload) platform. The linker is a protease-cleavable tripeptide linker, and the payload is a DNA TOP01i. YL205 is currently in a phase I/II clinical trial for solid tumors, including patients with locally advanced or metastatic EC (98).

Another NaPi2b-directed ADC is ZW220, which is a IgG1 mAb linked to ZD06519, a TOP01i payload and a camptothecin derivative (99, 100). Currently, ZW220 is being assessed in preclinical models of EC, as well as ovarian cancer and other malignancies (97).

CLDN6

CLDN6 function

Claudin 6 (CLDN6) is a four-pass transmembrane protein and a component of epithelial tight junctions, where it contributes to cell-cell adhesion, epithelial polarity, and regulation of paracellular permeability (101). CLDN6 is predominantly expressed during embryonic development and participates in epithelial differentiation and organogenesis, but its expression is largely silenced in healthy adult tissues. It is aberrantly re-expressed in several solid tumors, including ovarian, endometrial, and germ cell tumors (101–103). This oncofetal expression pattern, together with its accessibility on the tumor-cell surface and limited expression in normal adult tissues, makes CLDN6 an attractive target for ADCs and other targeted therapies (104, 105). Aberrant CLDN6 expression may also influence tumor-cell proliferation, migration, invasion, and intracellular signaling; however, these effects appear to be tumor type- and context-dependent (102).

CLDN6 expression and role in EC

In support of efforts to target CLDN6 in EC, a recent study evaluated CLDN6 membrane expression by IHC in tissue microarrays from a prospectively collected EC cohort comprising 1106 primary tumors and 187 metastatic lesions (106). CLDN6 expression was scored according to the Gastric criteria, yielding scores of 0, 1+, 2+, or 3+. Membrane staining was variable, ranging from negative to strong. For subsequent analyses, scores of 0 and 1+ were classified as low expression, whereas scores of 2+ and 3+ were classified as high expression. CLDN6 expression was detected in 18% of the EC cases evaluated, and high expression was observed in 10% of all samples. They observed high expression in 49% of USC and in 22% of carcinosarcomas. The authors further reported that high CLDN6 expression independently predicted poor disease-specific survival in a multivariate analysis (HR: 1.7, CI: 1.2–2.4, $p = 0.002$). Importantly, among patients whose primary tumors showed high CLDN6 expression, 77% retained high expression in at least one corresponding metastatic lesion, supporting CLDN6 as a potential ADC target in advanced EC.

CLDN6 ADCs in clinical trials (Table 5)

CLDN6-directed ADCs have entered early-phase clinical development. TORL-1-23, also known as ixotatug vedotin, is a CLDN6-targeted ADC containing a microtubule-disrupting payload and is being evaluated in a first-in-human phase I study in patients with CLDN6-positive advanced solid tumors, including EC and ovarian cancers (NCT05103683). Preliminary results have demonstrated encouraging antitumor activity across CLDN6-positive tumors; however, efficacy outcomes specific to EC have not yet been reported in detail.

DS-9606, a CLDN6-directed ADC incorporating a modified pyrrolobenzodiazepine payload, was also evaluated in a phase I basket trial that included a small number of patients with EC (NCT05394675), although no EC-specific efficacy signal was reported, and its clinical development was subsequently discontinued.

Nectin-4

Nectin-4 function

Nectin-4 is a membrane glycoprotein that is normally enriched in the placenta and is involved in Ca^{2+} -independent cell-cell adhesion and adherens junctions (107, 108). Its expression levels are elevated in several solid tumor types, including breast, colorectal, pancreatic, ovarian, and lung (109, 110). Overexpression of Nectin-4 is associated with increased cell proliferation, motility, and metastasis, epithelial-to-mesenchymal transition (EMT) signaling, and angiogenesis (107, 111).

Nectin-4 expression and role in EC

Few studies have investigated Nectin-4 expression in EC (111, 112). In a recent study, Chang et al. (111) examined 320 tissue samples from patients diagnosed with EC to determine Nectin-4 expression levels and their relationship to diagnosis and prognosis. Nectin-4 expression was significantly higher in EC tumor tissue compared to normal, healthy tissue. Furthermore, Nectin-4 expression was associated with a deficiency in DNA mismatch repair proteins. More revealing was that patients with MSH2/6-deficient EC who had high Nectin-4 expression had shorter PFS than those with low Nectin-4 expression (111).

Nectin-4 ADCs in clinical trials (Table 5)

There is currently one Nectin-4 ADC in a clinical trial including EC patients. Enfortumab vedotin (EV; ASG-22CE) is a Nectin-4-targeting ADC composed of a fully human immunoglobulin G1 monoclonal antibody conjugated to the microtubule-disrupting agent MMAE through a protease-cleavable maleimidocaproyl–valine-citrulline linker, with a DAR of approximately 4. EV is currently being evaluated in an open-label, single-arm phase II trial in patients with recurrent or persistent advanced or metastatic EC who have experienced disease progression after

prior chemotherapy and anti-PD-1 or other immunotherapy (NCT07139977). The study plans to enroll up to 12 patients and will assess the efficacy and safety of EV monotherapy. Clinical results have not yet been reported.

TF

TF function

Tissue factor (TF) is an important member of the coagulation cascade and is expressed on the surfaces of many tumor cells, including pancreatic, colon, gastric, esophageal, breast, prostate, kidney, and bladder cancers (113). Tumors often promote vascularization to increase access to oxygen and signaling factors that promote their growth; however, these new vessels are permeable and leaky (113). TF is expressed on tumor cell surfaces and activated by other coagulation factors (such as factor VII) that leak from those permeable vessels (113). Activation of TF can lead to thrombogenesis, tumor growth, mobility, adhesion, and immune evasion (113).

TF expression and role in EC

Expression of TF is generally associated with poor prognosis and thrombotic tendencies in many cancer types (113). In EC, TF is expressed in USC and is mostly highly expressed in the clear cell histology (113). In a study of 229 EC tumor tissues, TF expression correlated with worse OS and was inversely correlated with CD8 + T cell infiltration, while positively correlated with Treg cells infiltration (113).

TF ADCs in clinical trials

Tisotumab vedotin (TV; HuMax-TF-ADC) is a tissue factor (TF; CD142)-targeting ADC composed of a human monoclonal antibody conjugated to the microtubule-disrupting agent MMAE through a protease-cleavable maleimidocaproyl–valine-citrulline–p-aminobenzyl carbamate linker, with a DAR of approximately 4. TV was evaluated in the first-in-human, open-label, multicenter phase I/II innovaTV 201 trial (NCT02001623), which enrolled patients with relapsed, advanced, or metastatic solid tumors, including EC. Fourteen patients with EC were included in the dose-expansion cohort, and preliminary antitumor activity was observed across the evaluated tumor types.

TV was also investigated in a phase I/II safety and dose-expansion study involving patients with tissue factor-expressing solid tumors, including EC (NCT02552121). Another TF ADC currently in a phase I clinical trial (NCT04925284) is XB002, which comprises a 25A3 human TF monoclonal antibody conjugated to a zovodotin linker payload (114). The clinical trial is a dose-escalation and expansion study, which includes a cohort of EC patients with locally advanced, recurrent, or metastatic disease and progression that has been documented radiographically after their last systemic therapy. The trial was completed in March 2025; no results have been posted.

Clinical evidence specifically supporting TV in EC remains limited, and no phase III trial has established its efficacy in this population. Clinical trials evaluating TF-targeting ADCs in EC are summarized in Table 5.

AXL

AXL function

AXL is a receptor tyrosine kinase that has been found to be overexpressed in many cancer types, including breast, prostate, ovarian, and EC (115). It has one known ligand, growth arrest-specific 6 (GAS-6), which, when bound, induces autophosphorylation of AXL and downstream activation of MAPK/ERK, PI3K/AKT, NFκB, and STAT signaling pathways (115). Activation of AXL leads to increased cell proliferation, migration, invasion, adhesion, and cell survival via anti-apoptotic signaling and pro-inflammatory cytokine production (115).

AXL expression and role in EC

Using IHC for protein and RT-PCR for mRNA expression levels, studies have found that endometrial tumors exhibit higher expression levels of AXL compared to healthy controls and is associated with advanced stage and grade tumors (115, 116). Further, endometrial tumor expression of AXL and GAS-6 has been found to be higher in those patients who had a poor response to chemotherapy compared to those who had a good response (117).

AXL ADCs in clinical trials

Enapotamab vedotin (EnaV; HuMax-AXL-ADC) is an AXL-targeting ADC composed of a human immunoglobulin G1 monoclonal antibody conjugated to the microtubule-disrupting agent MMAE through a protease-cleavable valine-citrulline linker, with a DAR of approximately 4 (118). EnaV was evaluated in a first-in-human, open-label, multicenter phase I/II trial involving patients with relapsed or refractory solid tumors, including EC (NCT02988817)(119). The trial comprised a dose-escalation phase followed by tumor-specific and biomarker-selected expansion cohorts. Five patients with EC were enrolled in the dose-escalation phase; however, EC-specific efficacy was not established because of the

limited number of patients. Overall, the clinical activity observed across the study cohorts was insufficient to support further development of EnaV. The clinical evaluation of AXL-targeting ADCs in EC is summarized in Table 5.

Table 5
Other antibody-drug conjugates evaluated in endometrial cancer

ADC	Target	Payload	Linker	DAR	Trial (NCT#)	Study population	Phase
HWK-016	MUC16/CA-125	CPT116 a novel topoisomerase I inhibitor	Proprietary cleavable linker using carbon-bridge cysteine-rebridging conjugation technology	6	HWK-016-101 (NCT07470853)	Basket (solid tumors; EC/ovarian expansion)	I
ZW220	NaPi2b/SLC34A2	ZD06519 a camptothecin-derived topoisomerase I inhibitor	Cleavable MC–GGFG–aminomethyl peptide linker	ND		Preclinical	Preclinical
YL205	NaPi2b/SLC34A2	Novel DNA topoisomerase I inhibitor	Proprietary TMALIN (Tumor Microenvironment Activable LINKer-payload) protease-cleavable tripeptide linker	8	NCT06459973	Basket (solid tumors)	I/II
Ixotatug vedotin (TORL-1-23)	CLDN6	Monomethyl auristatin E (MMAE) a microtubule-disrupting agent	Protease-cleavable valine-citrulline linker	4	TORL123-001/TRIO-049 (NCT05103683)	Basket (solid tumors)	I
Enfortumab vedotin (EV; ASG-22CE)	Nectin-4	Monomethyl auristatin E (MMAE) a microtubule-disrupting agent	Protease-cleavable maleimidocaproyl–valine-citrulline linker	4	NCT07139977	EC-focused	II
Tisotumab vedotin (TV; HuMax-TF-ADC)	Tissue factor/CD142	Monomethyl auristatin E (MMAE) a microtubule-disrupting agent	Protease-cleavable mc–valine-citrulline–PABC linker	~ 4	NCT03245736	EC-focused	II
					innovaTV 201 (NCT02001623)	Basket (solid tumors)	I/II
					innovaTV 202 (NCT02552121)	Basket (solid tumors)	I/II
XB002	Tissue factor/CD142	ZD02044/N-acyl sulfonamide auristatin (MMAE) a microtubule-disrupting agent	Zovodotin (Protease-sensitive maleimidotrioxadodecanoyl-valine-citrulline linker)	~ 3.8	NCT04925284	Basket (solid tumors)	I

Dual targeting ADCs

The efficacy of conventional single-target ADCs may be limited by heterogeneous or reduced target-antigen expression, which has been supported by clinical and translational studies. Altered ADC internalization and intracellular trafficking have also been identified as potential resistance mechanisms, although the evidence for these processes is derived predominantly from preclinical models (120, 121). Although bystander activity may partially mitigate antigen heterogeneity, its effectiveness depends on sufficient ADC uptake and payload release from

antigen-positive cells, payload membrane permeability, and the spatial proximity of antigen-positive and antigen-negative tumor cells. It may therefore be insufficient in tumors with very low, spatially segregated, or completely lost target-antigen expression.

Dual-targeting, or bispecific, ADCs are designed to recognize two distinct tumor-associated antigens, thereby potentially expanding the population of targetable tumor cells and improving antibody avidity, internalization, and intracellular payload delivery (122). These agents should be distinguished from biparatopic ADCs, which recognize two non-overlapping epitopes on the same antigen, and from dual-payload ADCs, which deliver two different cytotoxic agents (123, 124) (Fig. 3).

Dual-targeting ADCs may broaden tumor-cell coverage and reduce escape driven by heterogeneous or lost expression of a single antigen; however, their efficacy depends on the expression patterns and spatial distributions of both targets, and their more complex molecular design may complicate manufacturing and pharmacokinetics (122, 125). Biparatopic ADCs may enhance avidity, receptor clustering, and internalization by engaging two epitopes on the same antigen, but they remain dependent on the expression of that single antigen and may not overcome complete antigen loss (126). Dual-payload ADCs may address payload-specific resistance and intratumoral heterogeneity by delivering cytotoxic agents with complementary mechanisms of action; however, controlling the payload ratio, linker stability, pharmacokinetics, and cumulative toxicity presents substantial development and manufacturing challenges (124, 127).

Among dual-targeting ADCs, ARR-002, also known as AV-P138-ADC, is a dual-targeting ADC being developed for EC and ovarian cancer. ARR-002 simultaneously targets MUC16 and NaPi2b using a tetravalent 2 + 2 antibody format to broaden tumor-cell coverage across heterogeneous antigen expression in EC and is site-specifically conjugated to the microtubule-disrupting payload MMAE through a valine-citrulline linker, with a DAR of 4 (128). Preclinical studies demonstrated binding to either antigen individually, simultaneous engagement of both targets, and enhanced internalization compared with single-target antibody controls (128). ARR-002 also showed greater antitumor activity than comparator single-target ADCs in an ovarian cancer xenograft model. An investigational new drug application for ARR-002 was cleared by the US Food and Drug Administration in May 2026, with initial clinical development focused on endometrial and ovarian cancer.

Although dual-targeting ADCs offer a rational strategy to broaden tumor-cell coverage and overcome antigen heterogeneity, their clinical benefit has not yet been established in EC (122, 123). Furthermore, dual-antigen binding does not necessarily confer true tumor-specific AND-gate activity, in which two distinct tumor-associated conditions must be simultaneously met before therapeutic activation, because some bispecific constructs can bind and internalize through either antigen independently rather than requiring simultaneous recognition of both targets (129). Careful selection of complementary antigen pairs, characterization of antigen co-expression and spatial heterogeneity, and evaluation of target expression in normal tissues will therefore be essential for defining tumor selectivity and the therapeutic window of these agents (123, 130).

Current challenges

ADC resistance

As with other treatment strategies, there is a high probability that some level of drug resistance will develop over time. ADC-targeted approaches are not exempt, so practitioners will need to plan for alternatives to circumvent anticipated inherent or acquired resistance. Potential mechanisms of ADC resistance are numerous and can develop at any point in the ADC mechanism of action. Antigen recognition can lead to resistance due to tumor cell heterogeneity and alterations in the density of target antigens on the tumor cells (131). Further, tumor cells alter the regulation of receptor internalization and recycling and increase shedding or cleavage of antigens (132). Even if the ADC enters the cell, resistance can arise from diminished trafficking of the ADC to the lysosome or from altered lysosomal function that prevents release of the cytotoxic payload (133, 134). Resistance to the cytotoxic payload itself can arise from increased activity of drug efflux pumps and from breakdown of the payload (135). Finally, tumor cells can also become resistant by hyperactivation of pro-survival cell signaling pathways, or disruption of programmed cell death pathways (136). Outside of the target tumor cell itself, another potential mechanism of resistance could include masking of the antigen by the extracellular matrix, preventing antibody recognition of the epitope (137, 138).

Drug combination strategies

Combination strategies may enhance the efficacy and durability of ADC treatment by addressing tumor heterogeneity, payload resistance, and the immunosuppressive tumor microenvironment. ADC-mediated tumor-cell death can promote tumor-antigen release, dendritic-cell activation, and immunogenic cell death, providing a biological rationale for combining ADCs with ICIs (139–141). In EC, this approach has been evaluated in a phase II study of MIRV plus pembrolizumab in patients with recurrent FRα-expressing USC. Although preclinical studies provide a biological rationale for combining ADCs with ICIs, clinical evidence in EC remains limited. In a small study of 18 evaluable patients, the combination produced a confirmed ORR of 28% and a 6-month PFS rate of 24.4%. All confirmed responses occurred in patients without prior exposure to ICIs. However, the study was closed before reaching its planned enrollment, precluding definitive conclusions regarding synergy, the effect of prior immunotherapy, or the contribution of the immune microenvironment (58).

ADC–immunotherapy combinations are also being investigated in earlier treatment settings. The ongoing phase III DESTINY-Endometrial01 trial (NCT06989112) is comparing T-DXd plus pembrolizumab or rilvegostomig, a bispecific PD-1/TIGIT inhibitor, with carboplatin, paclitaxel, and pembrolizumab as first-line therapy for HER2-expressing, pMMR, primary advanced or recurrent EC. Other potential combination partners include antiangiogenic agents, DNA-damage response inhibitors, conventional chemotherapy, and agents that modulate ADC internalization, lysosomal processing, or payload sensitivity. Although these approaches may enhance tumor penetration, immune activation, or cytotoxicity and may delay the development of resistance, EC–specific evidence remains limited. Future studies should incorporate biomarker-guided patient selection and carefully evaluate treatment sequence, prior payload exposure, and overlapping toxicities.

ADC after ADC

As the number of available ADCs increases, sequential treatment with one ADC after progression on another is likely to become increasingly relevant. The activity of a subsequent ADC may depend on whether it changes the target antigen, payload class, linker, or other structural features relative to the preceding ADC (142).

Sequential use of ADCs directed against the same antigen but carrying different payloads may retain activity when resistance is driven primarily by the first payload, linker, or intracellular processing rather than by loss of the target antigen. The clearest clinical example is the use of T-DXd after trastuzumab emtansine in HER2-positive breast cancer (143, 144). Although both ADCs target HER2, trastuzumab emtansine contains the microtubule inhibitor DM1, whereas T-DXd carries a TOPO1i payload, has a cleavable linker and a bystander effect. In DESTINY-Breast02, T-DXd demonstrated substantial clinical benefit in patients previously treated with trastuzumab emtansine, supporting the feasibility of sequential ADCs targeting the same antigen when their payload and pharmacologic properties differ (143). Nevertheless, this approach may be ineffective if resistance is caused by reduced or heterogeneous antigen expression, impaired antibody binding, or defective internalization.

Switching to an ADC directed against a different antigen may overcome resistance caused by loss or downregulation of the original target. However, cross-resistance may persist when both ADCs use payloads with the same mechanism of action. This issue has been examined in two retrospective studies of HER2-low metastatic breast cancer using sequential treatment with the HER2-targeted ADC T-DXd and the TROP2-targeted ADC SG, both of which deliver a TOPO1i payload (145, 146). Some patients derived benefit from the second ADC; however, response rates and treatment duration are generally lower with the second agent than with the first, suggesting partial cross-resistance related to the shared payload class (145, 146). These findings indicate that changing the target antigen may not fully overcome resistance when sequential ADCs use payloads with the same mechanism of action.

The optimal sequence of ADCs remains undefined. Available retrospective data suggest that responses to a second ADC are possible, including after progression on the first, but the magnitude and duration of benefit are frequently reduced (145, 147). Selection of the subsequent ADC should therefore consider persistent expression of the target antigen, the mechanism of the first payload, prior toxicities, linker and conjugation characteristics, and the potential for overlapping resistance. When feasible, switching both the target and the payload class may offer the greatest theoretical opportunity to avoid cross-resistance, although this strategy has not yet been prospectively validated. Repeat tumor sampling or molecular profiling after ADC progression may help distinguish target-related resistance from payload-related resistance and inform subsequent treatment selection.

These considerations are particularly relevant to EC, in which multiple ADC targets—including HER2, FRα, TROP2, B7-H4, MUC16, NaPi2b, and CLDN6—may be expressed within the same tumor or across different tumor subclones and are currently being developed for use in EC. However, there are currently insufficient clinical data to establish an optimal ADC sequence in EC. Prospective studies should incorporate longitudinal assessment of antigen expression, prior payload exposure, and mechanisms of resistance when evaluating ADC-after-ADC strategies.

Biomarkers of response

Identification of predictive biomarkers will be essential for optimizing patient selection for ADC therapy. At present, target-antigen expression, generally assessed by IHC, is the most clinically established biomarker (148). However, antigen expression alone does not consistently predict treatment response, highlighting the need for composite biomarkers that integrate target abundance and heterogeneity with features related to ADC processing and payload sensitivity (148, 149).

In EC, HER2 expression has shown the clearest association with response: in DESTINY-PanTumor02, T-DXd demonstrated greater activity in tumors with HER2 IHC 3+ expression than in those with IHC 2+ expression, supporting HER2 protein level as an enrichment biomarker (45, 150). However, HER2 expression is often heterogeneous and may differ between primary and metastatic lesions, suggesting that assessment of recent or metastatic tissue, when available, may be more informative than testing archival primary tumors alone. FRα expression may similarly enrich for response to MIRV. In recurrent USC, responses were numerically more frequent in tumors with higher FRα expression, although FRα expression alone was not significantly associated with response or PFS (58). An exploratory composite biomarker incorporating

FR α expression and immune-microenvironment features showed a stronger association with response, suggesting that integrated biomarkers may outperform antigen expression alone, particularly for ADC–immunotherapy combinations (58).

Biomarkers beyond antigen expression that are clinically applicable are limited. For ADCs carrying TOP01i payloads, TOP1 abundance or enzymatic activity, SLFN11 expression, homologous-recombination deficiency, replication stress and other DNA-damage response pathways, and apoptotic competency may influence payload sensitivity (151). For ADCs carrying microtubule-disrupting payloads, tubulin composition and dynamics, microtubule-regulatory proteins, and multidrug-efflux transporters may be relevant (152). In addition, membrane-permeable payloads that can diffuse into neighboring antigen-negative cells and mediate a bystander effect, may enable activity in tumors with heterogeneous or relatively low target-antigen expression (153).

The tumor microenvironment may provide an additional layer of predictive information. Stromal density and extracellular matrix composition can restrict ADC penetration and intratumoral distribution (137). Tumor-associated macrophages can internalize and process ADCs through Fc γ receptor–mediated interactions, potentially contributing to intratumoral payload release. Conversely, Fc γ receptor–mediated uptake by myeloid cells may divert ADCs from tumor cells and contribute to nonproductive clearance (147, 148). Immune-cell infiltration, PD-L1 expression, and the capacity of the ADC payload to induce immunogenic cell death may be particularly relevant to response when ADCs are combined with ICIs (58, 140). However, no immune or tumor microenvironment biomarker has yet been prospectively validated to predict response to ADC-based therapy in EC.

Because target-antigen expression and payload sensitivity can evolve during disease progression and under treatment pressure, biomarker assessment should ideally be longitudinal rather than limited to the original diagnostic specimen. Repeat biopsy of recurrent or progressing disease and analysis of multiple tumor regions may help capture spatial and temporal heterogeneity. In EC, substantial discordance in HER2 expression has been observed between paired primary and metastatic lesions, supporting reassessment of the therapeutic target when clinically feasible (154). Longitudinal liquid-biopsy approaches, particularly circulating tumor DNA, may provide complementary information on clonal evolution, treatment response, and emerging resistance, while antigen-specific molecular imaging could enable noninvasive, whole-body assessment of target expression across multiple lesions (155). Prospective EC trials should therefore incorporate serial tissue and liquid-biopsy collection and explore molecular imaging to identify biomarkers associated with initial response, durability of benefit, and cross-resistance to subsequent ADCs.

Conclusions

ADCs are rapidly expanding the therapeutic landscape of EC by enabling selective delivery of highly potent cytotoxic payloads to tumor cells. HER2-directed ADCs currently provide the strongest clinical proof of concept, while agents targeting FR α , TROP2, B7-H4, MUC16, NaPi2b, and CLDN6 are broadening the range of potentially actionable antigens. Dual-targeting constructs, novel payloads and linkers, rational drug combinations, and sequential ADC strategies may further improve tumor coverage and help overcome resistance.

However, successful clinical implementation will require more than identification of target-antigen positivity. Antigen density, spatial and temporal heterogeneity, internalization and trafficking, payload sensitivity, the tumor microenvironment, prior treatment exposure, and overlapping toxicities may all influence efficacy. Prospective EC trials should therefore incorporate standardized antigen assessment, longitudinal tissue and liquid-biopsy analyses, and translational studies of resistance. Ultimately, biomarker-guided selection and rational sequencing of ADCs will be essential to translate the growing number of available agents into durable and individualized treatment strategies for patients with EC.

Abbreviations

ADCs, antibody-drug conjugates

AF-HPA, auristatin F-hydroxypropylamide

AKT, protein kinase B

ARR-002, AV-P138 antibody–drug conjugate

B7-H4, B7 homolog 4

CA-125, cancer antigen 125

CI, confidence interval

CLDN6, claudin 6

DAR, drug-to-antibody ratio

Dato-DXd, datopotamab deruxtecan

dMMR, mismatch repair-deficient

DM1, emtansine

DM4, ravtansine

EC, endometrial cancer

EMT, epithelial-to-mesenchymal transition

EnaV, enapotamab vedotin

ERBB, erythroblastic oncogene B

EV, enfortumab vedotin

FDA, Food and Drug Administration

FR α , folate receptor alpha

GAS6, growth arrest-specific 6

HER2, human epidermal growth factor receptor 2

HR, hazard ratio

ICI, immune checkpoint inhibitor

IgG1, immunoglobulin G1

IHC, immunohistochemistry

MAPK, mitogen-activated protein kinase

MIRV, mirvetuximab soravtansine

MMAE, monomethyl auristatin E

Mo-Rez, mocertatug rezetecan

MUC16, mucin 16

NaPi2b, sodium-dependent phosphate transport protein 2b

NSMP, no specific molecular profile

ORR, objective response rate

OS, overall survival

PABC, p-aminobenzyl carbamate

PD-1, programmed cell death protein 1

PD-L1, programmed death-ligand 1

PFS, progression-free survival

pMMR, mismatch repair-proficient

POLE, DNA polymerase epsilon

PROC, platinum-resistant ovarian cancer

Rina-S, rinatabart sesutecan

RT-PCR, reverse transcription polymerase chain reaction

sac-TMT, sacituzumab tirumotecan

SG, sacituzumab govitecan

T-DM1, ado-trastuzumab emtansine

T-DXd, trastuzumab deruxtecan

TF, tissue factor

TIGIT, T-cell immunoreceptor with immunoglobulin and ITIM domains

TOP2 α , topoisomerase II inhibitor

Treg, regulatory T cell

TROP2, trophoblast cell-surface antigen 2

TV, tisotumab vedotin

UCS, uterine carcinosarcoma

USC, uterine serous carcinoma

Declarations

Conflict of interest:

S.B. reports consulting fees from ImmunoGen, Speaker fees from OncLive

Competing Interests

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Authors' Contributions

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Figures

Basic components of an ADC

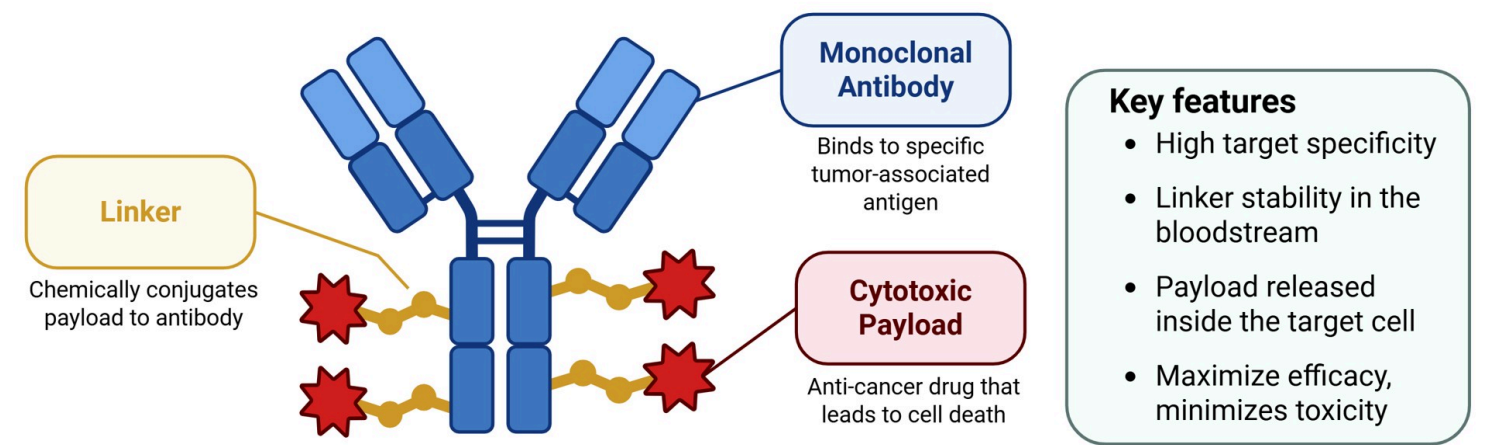


Figure 1

Basic components of an antibody–drug conjugate (ADC).

An ADC consists of three principal components: a monoclonal antibody, a chemical linker, and a cytotoxic payload. The monoclonal antibody selectively binds to a tumor-associated antigen expressed on the surface of cancer cells. The linker connects the payload to the antibody. Once the payload is released, it induces cell death of the target cell. The coordinated design of these components aims to maximize antitumor efficacy while minimizing systemic toxicity. Created in BioRender. Rodman, E. (2026) <https://BioRender.com/0smtpye>.

ADC mechanism of action

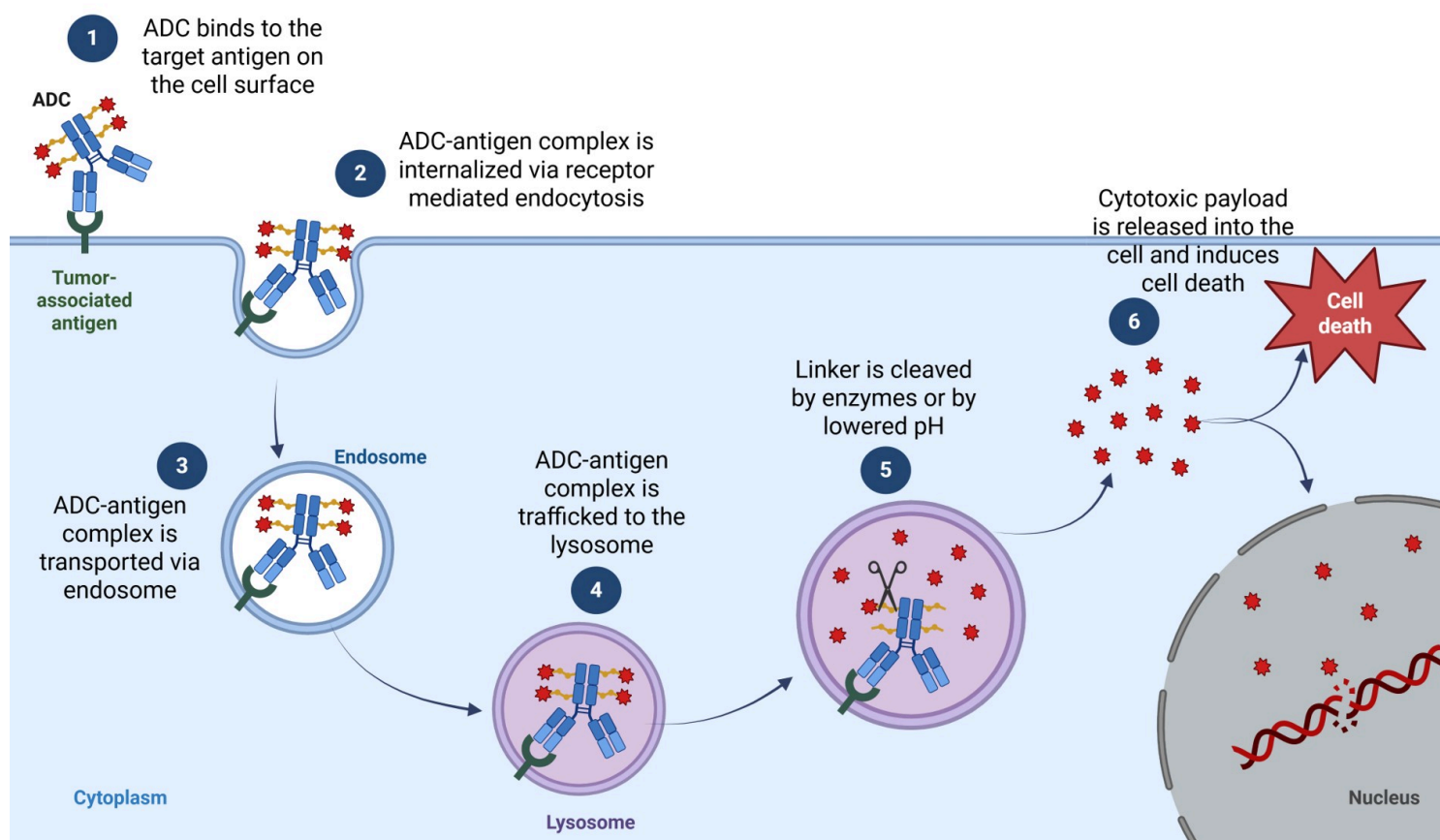


Figure 2

Antibody–drug conjugate (ADC): Mechanism of action.

After binding to a tumor-associated antigen on the cancer cell surface, the ADC–antigen complex is internalized through receptor-mediated endocytosis and transported within an endosome. The complex is subsequently trafficked to the lysosome, where acidic conditions and lysosomal enzymes promote linker cleavage and/or antibody degradation, resulting in intracellular release of the cytotoxic payload. The released payload then interacts with its intracellular target, inducing DNA damage and/or disrupting essential cellular processes, ultimately leading to tumor cell death. Created in BioRender. Rodman, E. (2026) <https://BioRender.com/s2kjr6l>.

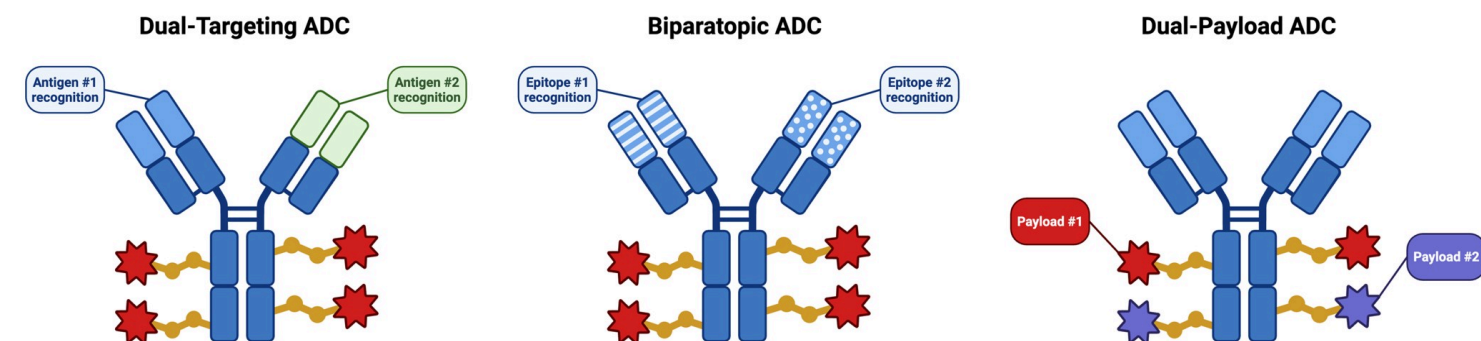


Figure 3

Advanced types of antibody-drug conjugates.

Dual-Targeting ADCs recognize two distinct tumor-associated antigens. Biparatopic ADCs recognize the same antigen but are designed for two non-overlapping epitopes of that antigen. Dual-Payload ADCs release two different types of cytotoxic payloads. Created in BioRender. Rodman, E. (2026) <https://BioRender.com/aflkfib>

Supplementary Files

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- [Table1HER2ver2withstudypopulation.xlsx](#)
- [Table2FRaver2withstudypopulation.xlsx](#)
- [Table3TROP2ver2withstudypopulation.xlsx](#)
- [Table4B7H4ver2withstudypopulation.xlsx](#)
- [Table5othersver2withstudypopulation.xlsx](#)